Valence Alignment in Human-AI Story Co-Creation

Analysis of Field Data and AI-AI Baseline Comparison

PENPAL Project

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Contents

0.1 Summary . . . 2

1 1. Data Loading and Preprocessing . . . 2
1.1 1.1 Load Data . . . 2
1.2 1.2 Data Preprocessing . . . 2
1.3 1.3 Compute Lagged Variables and Cosine . . . 3

2 2. Descriptive Statistics . . . 3
2.1 2.1 Token Count Statistics . . . 3
2.2 2.2 Valence Distribution Statistics . . . 3

3 3. Valence Distribution Analysis . . . 3
3.1 3.1 Valence Distribution Difference Test . . . 3
3.2 3.2 Combined Valence Distribution Plot . . . 4

4 4. Valence Alignment Analysis . . . 5
4.1 4.1 Per-Story Correlation (Same Turn) . . . 5
4.2 4.2 Per-Story Correlation (Previous Turn) . . . 5
4.3 4.3.1 All turns story correlation (same turn) . . . 5
4.4 4.3.2 All turns story correlation (Previous Turn) . . . 5
4.5 4.3.3 Probe Sentiment Narrative Development . . . 6
4.6 4.4.1 Alignment Comparison Plot . . . 7
4.7 4.4.2 Comparison: Per-Story vs Aggregate Correlations . . . 7
4.8 4.6 Statistical Tests: Alignment Differences . . . 8
4.9 Cosine Distribution Difference Test . . . 9
4.10 Valence Distance Difference Test . . . 10

5 5. Summary and Conclusions . . . 11
5.1 5.1 Key Findings Summary . . . 11

6 6. Supplementary: Interaction Effects . . . 11
6.1 6.1 Starter Interaction Effects . . . 11
6.1.1 6.1.1 Same-Turn Alignment by Starter . . . 11
6.1.2 6.1.2 Previous-Turn Alignment by Starter . . . 12
6.1.3 6.1.3 Per-Story Correlations by Starter . . . 12
6.2 6.2 LLM Type Interaction Effects . . . 12
6.2.1 6.2.1 Same-Turn Alignment by LLM Type . . . 12
6.2.2 6.2.2 Previous-Turn Alignment by LLM Type . . . 13
6.2.3 6.2.3 Per-Story Correlations by LLM Type . . . 14

6.3 6.3 Valence Distribution by Conditions . . . 14
6.3.1 6.3.1 Valence by Starter . . . 14
6.3.2 6.3.2 Valence by LLM Type . . . 15
6.4 7. Session Information . . . 15

0.1 Summary

- Emotional alignment is asymmetric: the model reliably aligns to the user (User → AI), while humans do not mirror the preceding AI tone.
- AI–AI interactions show higher, bidirectional alignment driven by symmetric context access.
- Field data shows greater variance in valence and weaker overall coordination compared to the simulated dataset.
- Participant-level modelling reveals a consistent "rubber-band" effect, where early divergence returns toward baseline in later stages.
- Overall, affective dynamics in the field are primarily model-driven rather than mutually convergent.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 820 16
```

```
## Simulated data dimensions: 30 12
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"                 "user"
##  [3] "ai"                        "conversation_id"
##  [5] "respondent_id"             "interaction_count"
##  [7] "llm_type"                  "turn"
##  [9] "starter"                   "user_embedding"
## [11] "ai_embedding"              "pct_turn"
## [13] "user_sentiment_score"      "ai_sentiment_score"
## [15] "user_sentiment_projection" "ai_sentiment_projection"
```

```
##
## Simulated data columns:
```

```
##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              81     648                    8           -0.0970        -0.00369
```

```
##
## === Simulated Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===
```

```
## Field data - Turn 1 with non-NA lag (should be 0): 0
```

```
## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

2 2. Descriptive Statistics

2.1 2.1 Token Count Statistics

```
## === Field Data Token Statistics ===
```

```
## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             25.0           5.46           29.0         7.78            0.900
```

```
##
## === Simulated Data Token Statistics ===
```

```
## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             26.0           2.17           41.1         3.73            0.636
```

2.2 2.2 Valence Distribution Statistics

```
## # A tibble: 4 x 5
##   data_source agent mean_valence sd_valence median_valence
##   <chr>       <chr>        <dbl>      <dbl>          <dbl>
## 1 Field       ai        -0.00369      0.391        -0.0212
## 2 Field       user      -0.0970       0.397        -0.121
## 3 Simulated   ai        -0.296        0.681        -0.487
## 4 Simulated   user      -0.407        0.529        -0.504
```

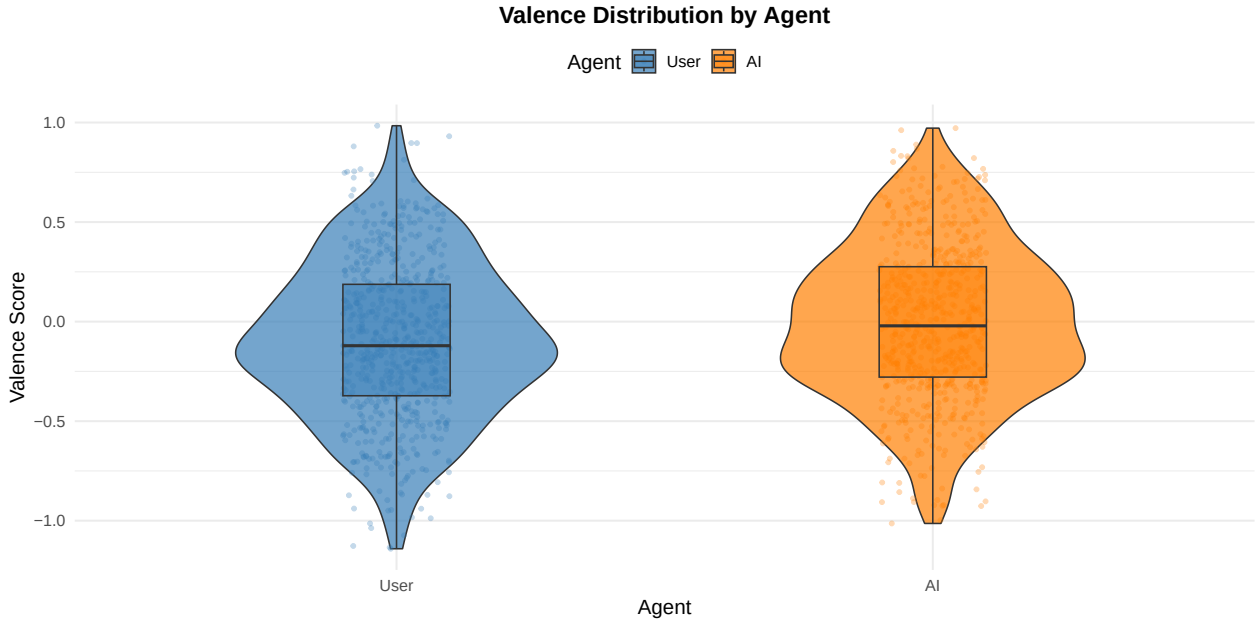
3 3. Valence Distribution Analysis

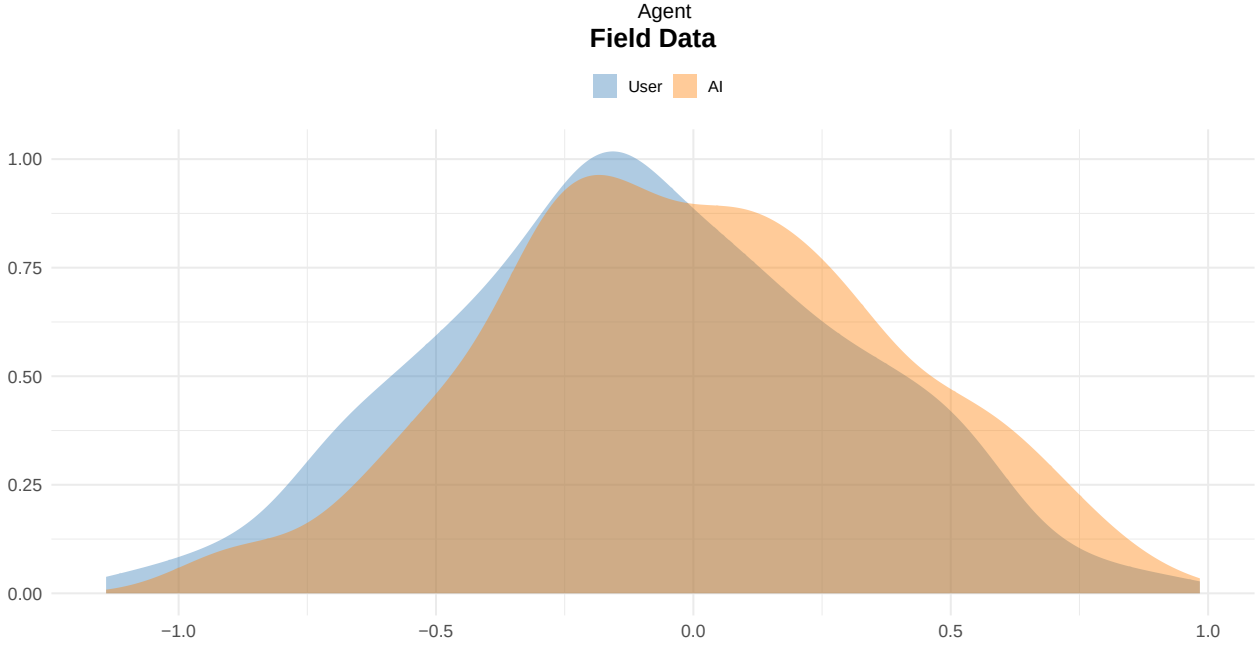
3.1 3.1 Valence Distribution Difference Test

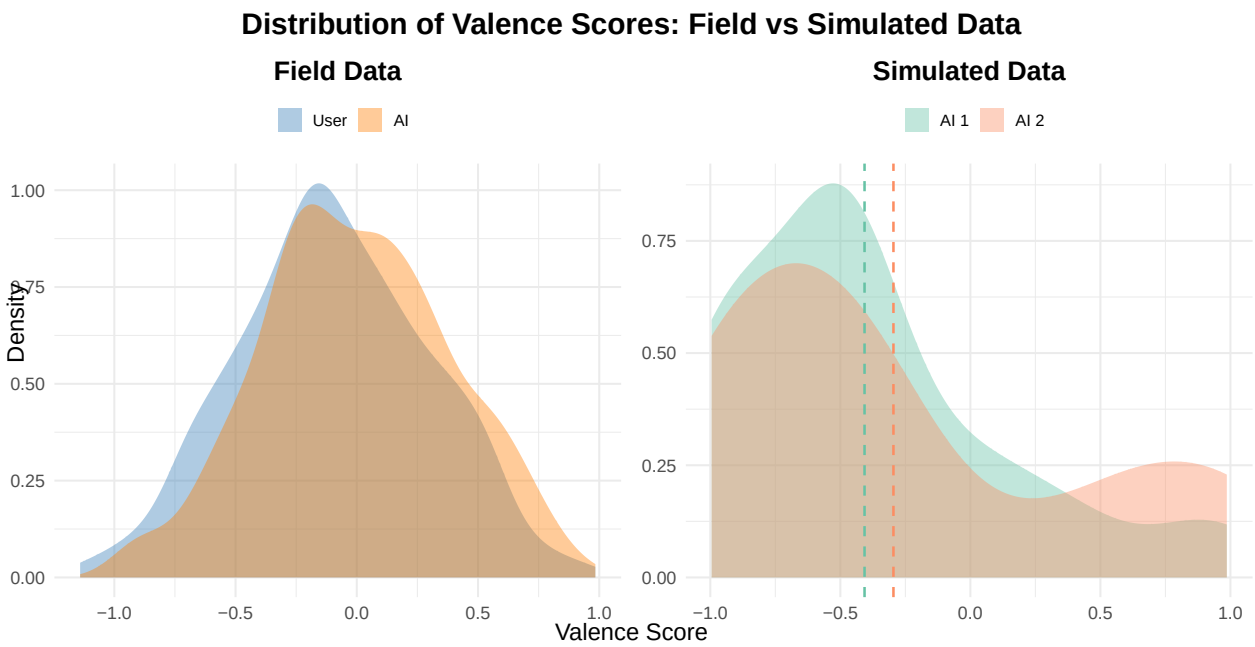
```
## === Valence Difference Model Coefficients ===
```

```
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.0970     0.0256 79.9995 -3.7832    3e-04
## agentAI       0.0933     0.0207 79.9993  4.4981    0e+00
```

3.2 3.2 Combined Valence Distribution Plot







4 4. Valence Alignment Analysis

4.1 4.1 Per-Story Correlation (Same Turn)

```
## === Same-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.211  0.384      0.206        81
## 2 Simulated     -0.246  0.440     -0.314         6
```

4.2 4.2 Per-Story Correlation (Previous Turn)

```
## === Previous-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field         0.0599  0.372      0.111        81
## 2 Simulated     0.533   0.224      0.546         6
```

4.3 4.3.1 All turns story correlation (same turn)

```
## === Same-Turn All Turns Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.411     NA      0.411         1
## 2 Simulated     -0.110     NA     -0.110         1
```

4.4 4.3.2 All turns story correlation (Previous Turn)

```
## === Previous-Turn Aggregate Correlation ===
```

```
## Field: Turn 1 observations (excluded): 0

## Field: Valid lag observations: 567

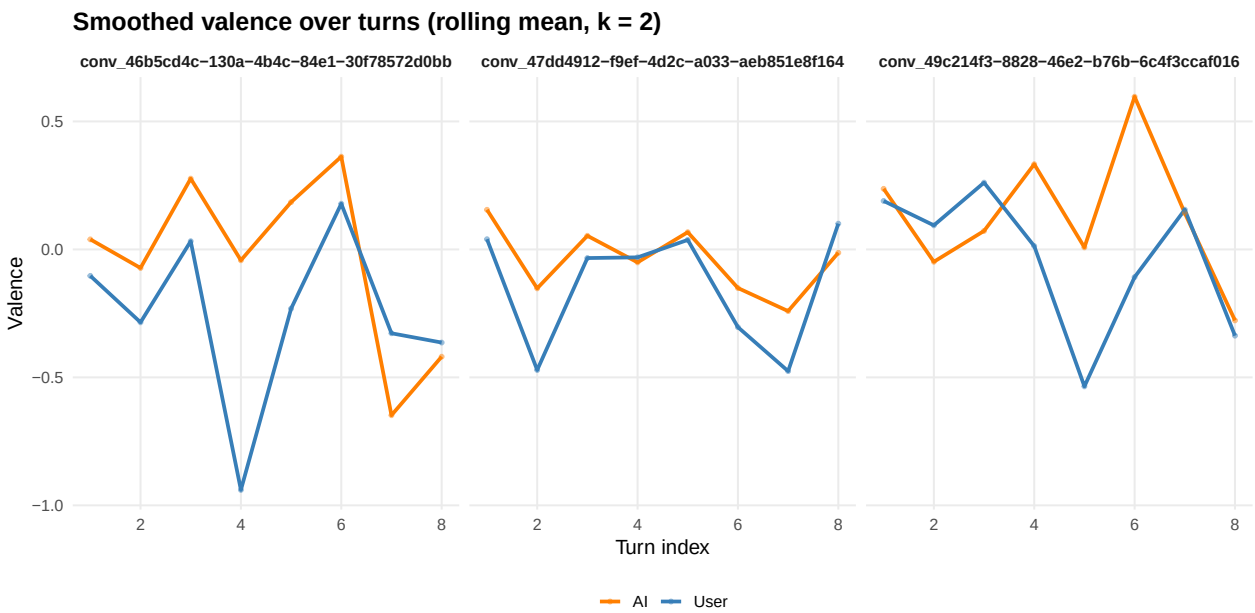
## Simulated: Turn 1 observations (excluded): 6

## Simulated: Valid lag observations: 24

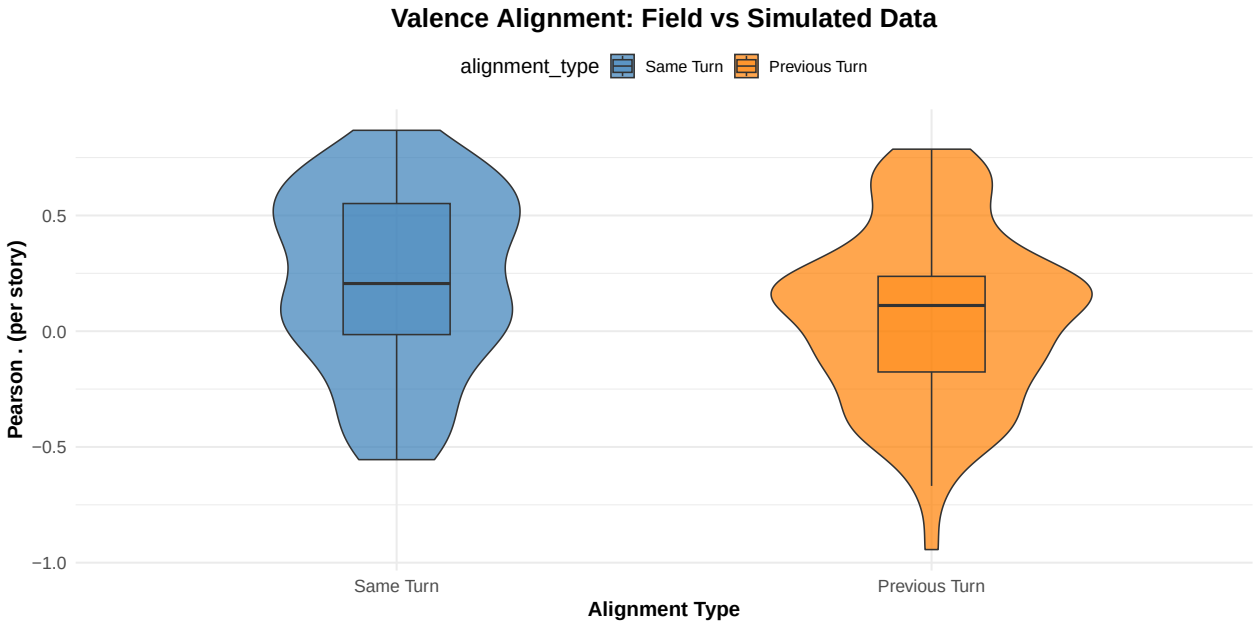
## === Previous-Turn All Turns Alignment Summary ===

## # A tibble: 2 x 6
##   data_source mean_rho sd_rho median_rho n_total n_valid
##   <chr>          <dbl>  <dbl>      <dbl>   <int>   <int>
## 1 Field          0.343     NA      0.343     648     567
## 2 Simulated      0.425     NA      0.425      30      24
```

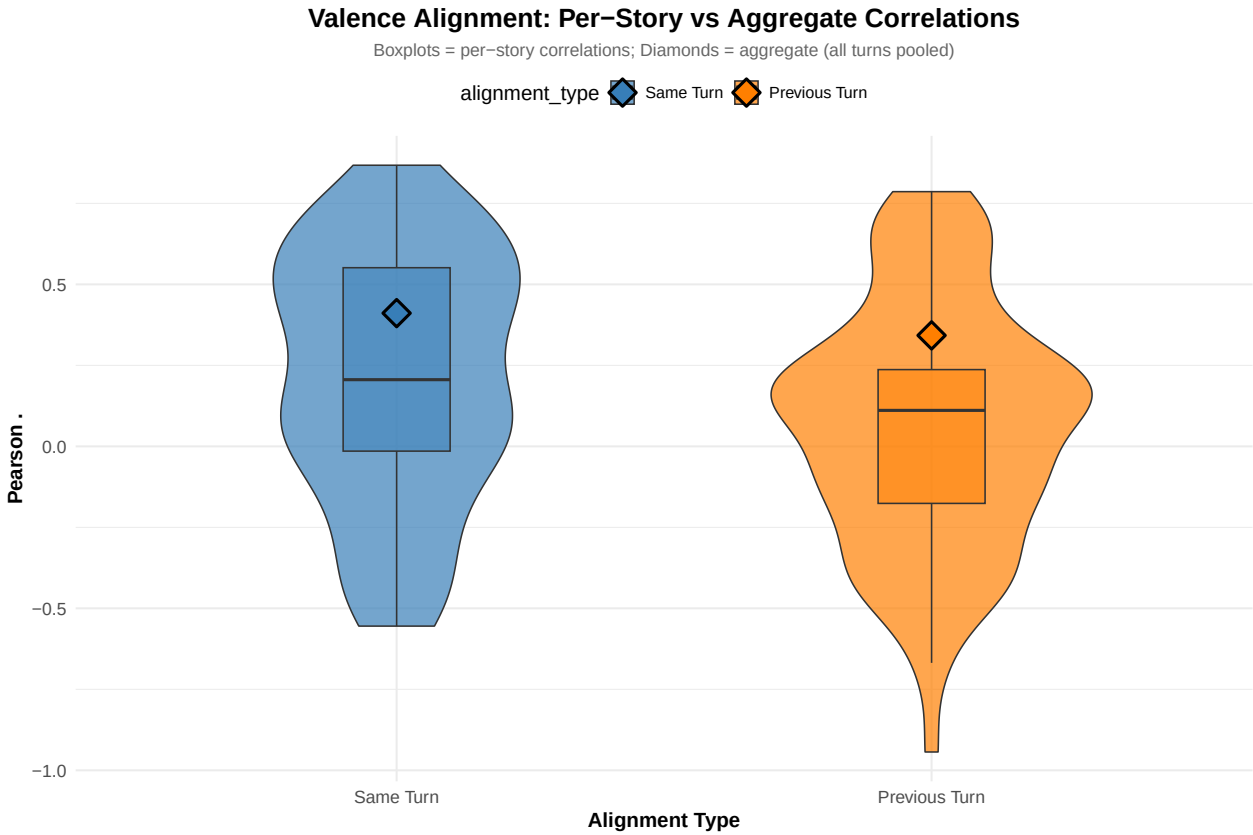
4.5 4.3.3 Probe Sentiment Narrative Development



4.6 4.4.1 Alignment Comparison Plot



4.7 4.4.2 Comparison: Per-Story vs Aggregate Correlations




```
##
## === Per-Story vs Aggregate Correlation Comparison ===
## # A tibble: 4 x 4
```

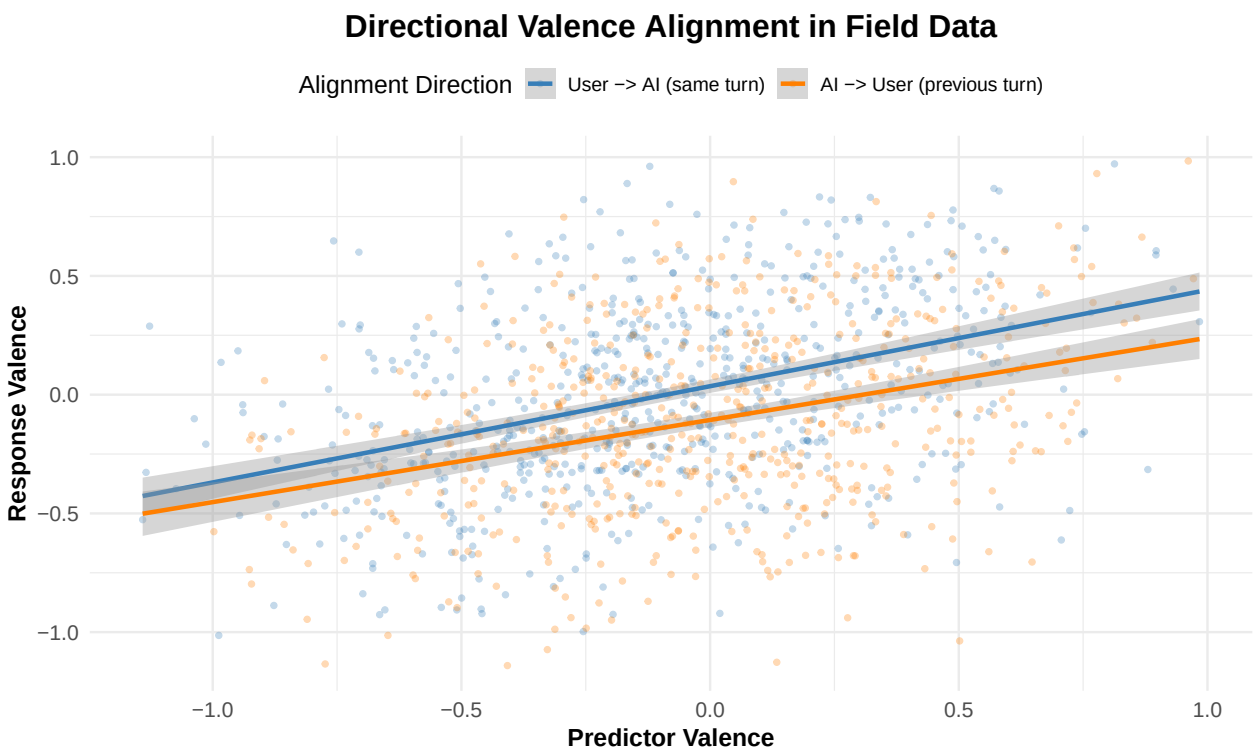
```
##   data_source alignment_type `Per-Story Mean` `Aggregate All Turns`
##   <chr>       <fct>                    <dbl>                 <dbl>
## 1 Field       Same Turn               0.211                  0.411
## 2 Field       Previous Turn           0.0599                 0.343
## 3 Simulated   Same Turn              -0.246                 NA
## 4 Simulated   Previous Turn           0.533                 NA
```

```
## === Directional Alignment Long-Format Data ===
```

```
## # A tibble: 2 x 5
##   alignment_direction n_rows mean_predictor mean_response n_conversations
##   <fct>               <int>          <dbl>         <dbl>           <int>
## 1 User_to_AI            648        -0.0970      -0.00369              81
## 2 AI_to_User            567        -0.0126      -0.111                81
```

```
## === Mixed Effects Model (Standardized): Directional Alignment (User -> AI vs AI -> User) ===
```

```
##                                          Estimate Std. Error   df t value
## (Intercept)                                0.0000     0.0362 1211  0.0000
## predictor_z                                0.2112     0.0387 1211  5.4633
## alignment_directionAI_to_User              0.0000     0.0529 1211  0.0000
## predictor_z:alignment_directionAI_to_User -0.1513     0.0569 1211 -2.6587
##                                          Pr(>|t|)
## (Intercept)                                1.0000
## predictor_z                                0.0000
## alignment_directionAI_to_User              1.0000
## predictor_z:alignment_directionAI_to_User  0.0079
```



4.8 4.6 Statistical Tests: Alignment Differences

```
##
## === One-Sample Tests Against Zero ===
```

```
## # A tibble: 4 x 6
##   data_source alignment_type t_stat    p_value mean_rho     n
##   <chr>       <fct>           <dbl>      <dbl>    <dbl> <int>
## 1 Field       Same Turn        4.95 0.00000403   0.211     81
## 2 Field       Previous Turn    1.45 0.151        0.0599    81
## 3 Simulated   Same Turn       -1.37 0.229       -0.246      6
## 4 Simulated   Previous Turn    5.84 0.00209      0.533      6
```

4.9 Cosine Distribution Difference Test

```
##
## === Cosine Sim: User (Self vs Interlocutor) ===

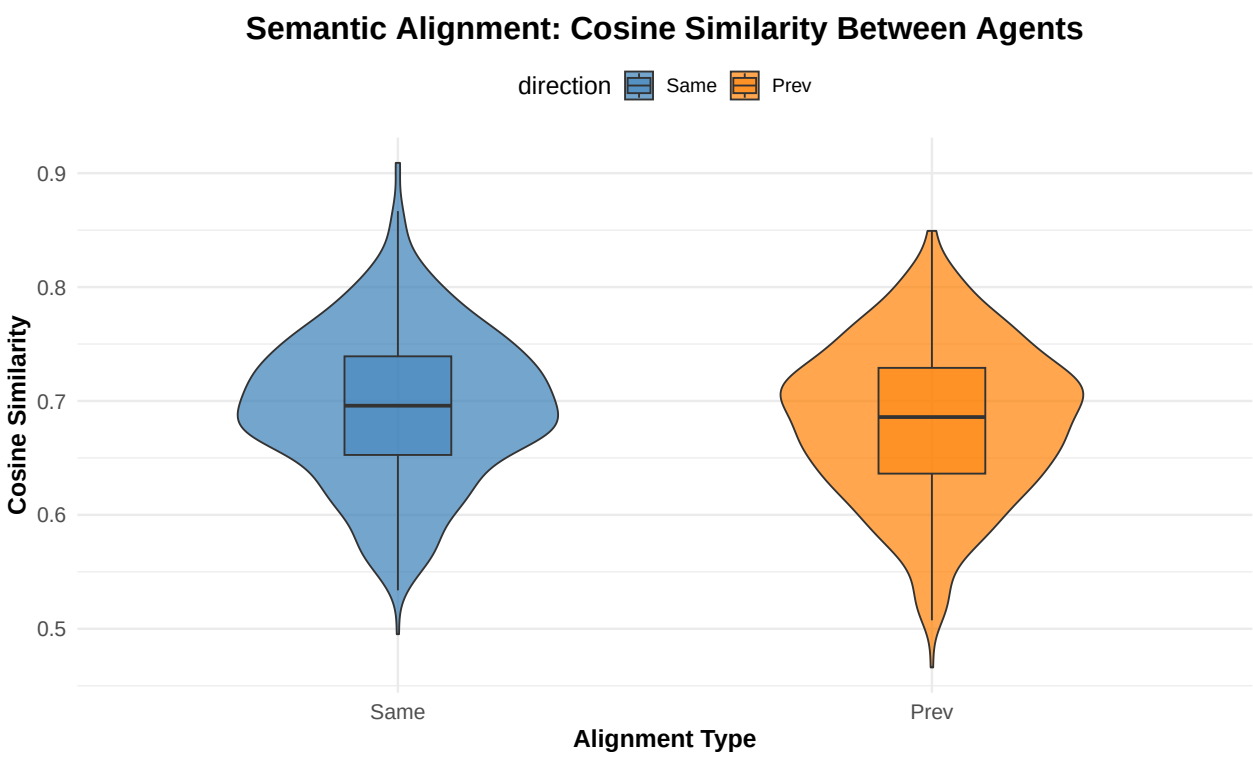
##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6976     0.0043 80.0014 163.0033    0e+00
## directionInterlocutor  -0.0157     0.0042 80.0009  -3.7100    4e-04

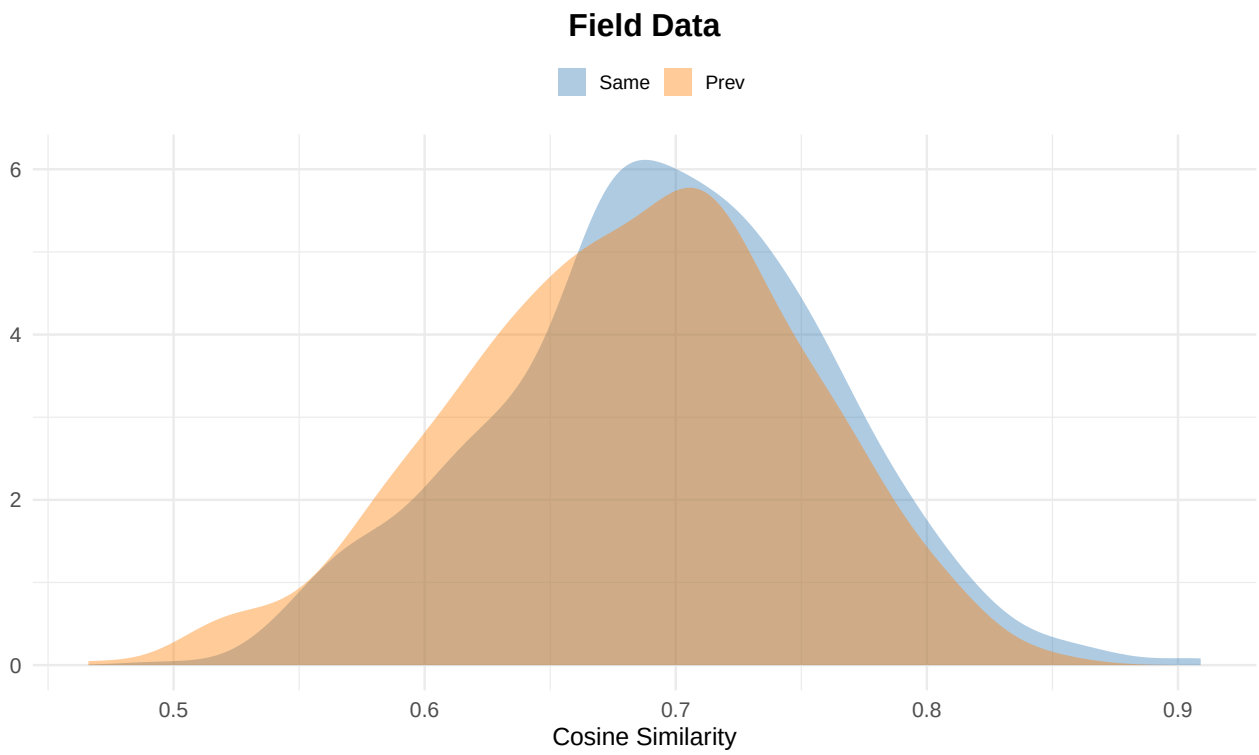
##
## === Cosine Sim: AI (Self vs Interlocutor) ===

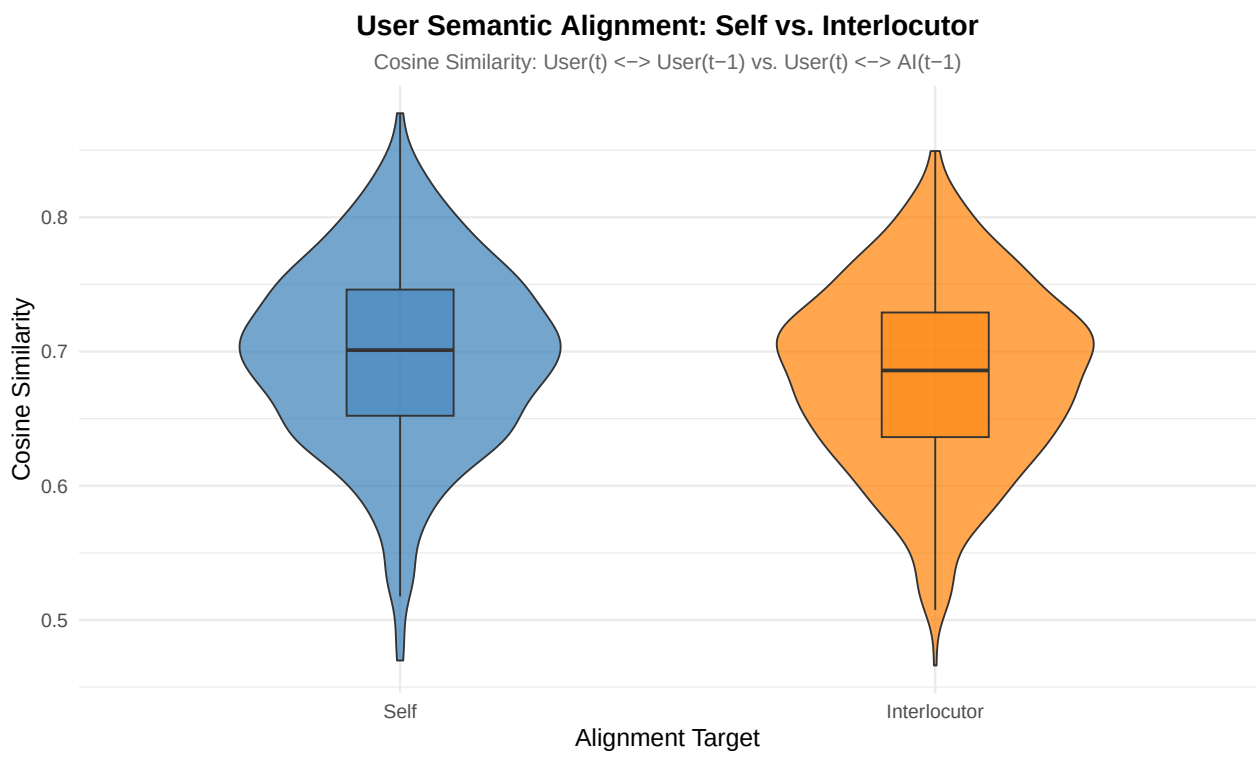
##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6859     0.0043 80.0001 159.3493   0.0000
## directionInterlocutor   0.0091     0.0036 80.0046   2.4889   0.0149

##
## === Cosine Sim: General Direction Comparison ===

##               Estimate Std. Error        df  t value Pr(>|t|)
## (Intercept)     0.6950     0.0039   84.6718 178.1176    0e+00
## directionPrev  -0.0131     0.0035 1110.1264  -3.7575    2e-04
```







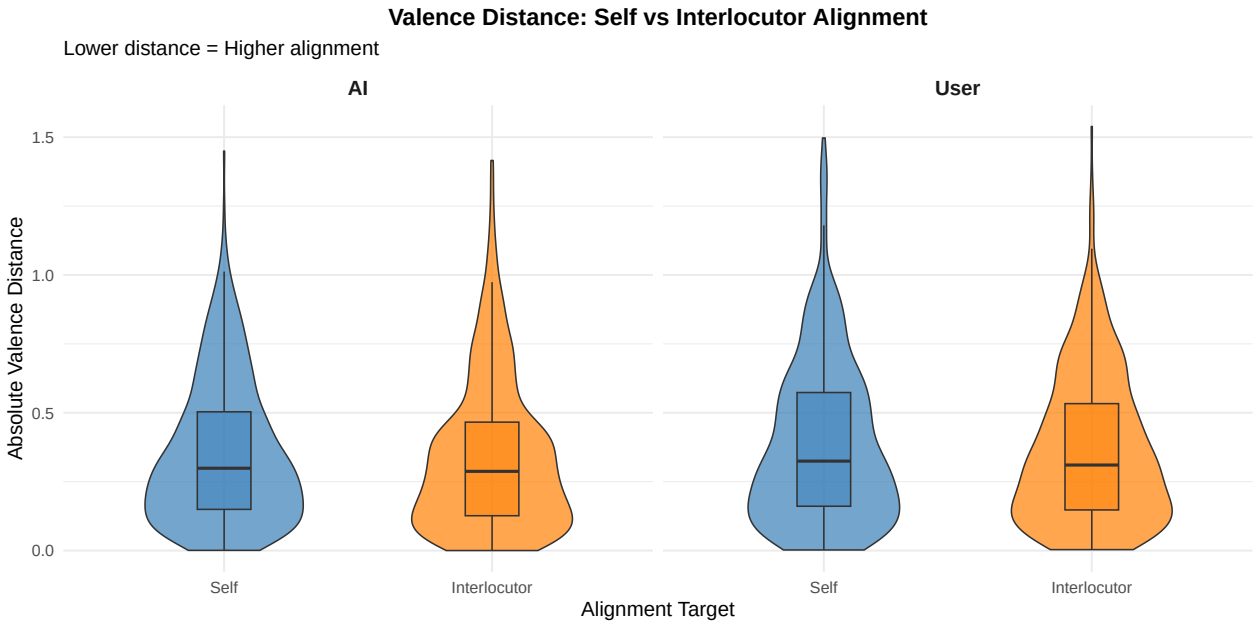
4.10 Valence Distance Difference Test

```
##
## === Valence Distance Model: User (Self vs Interlocutor) ===
##                        Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept)             0.3877     0.0150  87.5721 25.7695   0.0000
## directionInterlocutor  -0.0199     0.0163 512.1857 -1.2217   0.2224
```

```
##
## === Valence Distance Model: AI (Self vs Interlocutor) ===
```

```
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             0.3551     0.0110 1053.8787 32.3385   0.0000
## directionInterlocutor  -0.0130     0.0166   91.3748 -0.7798   0.4376
```



5 5. Summary and Conclusions

5.1 5.1 Key Findings Summary

```
## # A tibble: 5 x 3
##   Metric                      Field_Data Simulated_Data
##   <chr>                            <dbl>          <dbl>
## 1 N Stories                       81              6
## 2 Mean User Valence               -0.097         -0.407
## 3 Mean AI Valence                 -0.004         -0.296
## 4 Same-Turn Alignment ( )          0.211         -0.246
## 5 Previous-Turn Alignment ( )      0.06           0.533
```

6 6. Supplementary: Interaction Effects

6.1 6.1 Starter Interaction Effects

6.1.1 6.1.1 Same-Turn Alignment by Starter

```
##             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  -0.0442     0.0339  240.3806 -1.3036   0.1936
## turn         -0.0011     0.0041 1214.0000 -0.2746   0.7837
```

```
## === Same-Turn Alignment: Starter Interaction ===
```

```
##                           Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept)                0.0400     0.0360  74.1364  1.1103   0.2705
## user_valence               0.3005     0.0490 628.8624  6.1334   0.0000
## starteruser               -0.0390     0.0513  74.5280 -0.7601   0.4496
## user_valence:starteruser  -0.0902     0.0698 640.5545 -1.2925   0.1967
```

6.1.2 6.1.2 Previous-Turn Alignment by Starter

```
## === Previous-Turn Alignment: Starter Interaction ===

##                              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                   -0.1100     0.0318  67.2238 -3.4592   0.0009
## ai_valence_lag1                0.1634     0.0599 538.3659  2.7251   0.0066
## starteruser                    0.0067     0.0453  67.3929  0.1472   0.8834
## ai_valence_lag1:starteruser    0.1383     0.0853 512.2713  1.6215   0.1055
```

6.1.3 6.1.3 Per-Story Correlations by Starter

```
## === Same-Turn Correlations by Starter ===

## # A tibble: 2 x 4
##   starter mean_rho sd_rho n_stories
##   <chr>      <dbl>  <dbl>     <int>
## 1 ai        0.343   0.379        41
## 2 user      0.0756  0.344        40

##
## T-test (starter effect on same-turn rho):

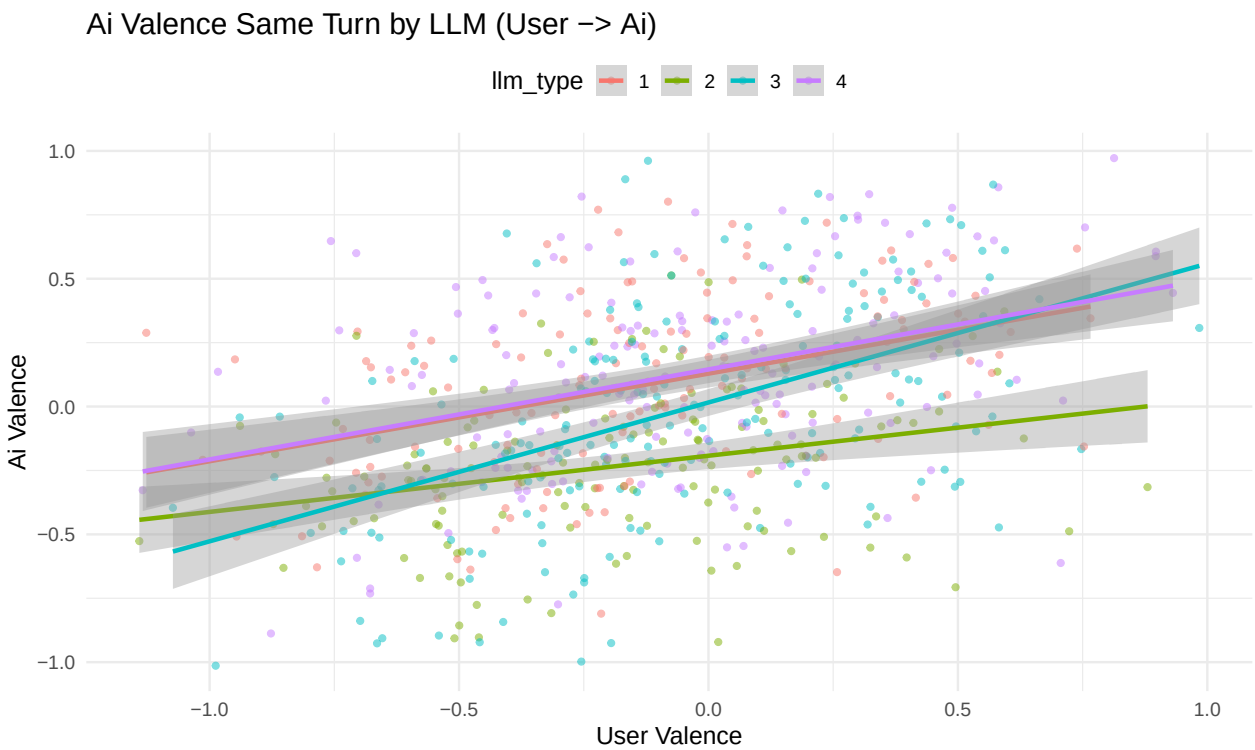
##
##  Welch Two Sample t-test
##
## data:  rho by starter
## t = 3.3355, df = 78.601, p-value = 0.001302
## alternative hypothesis: true difference in means between group ai and group user is not equal to 0
## 95 percent confidence interval:
##  0.1079888 0.4276704
## sample estimates:
##   mean in group ai mean in group user
##         0.34344577         0.07561613
```

6.2 6.2 LLM Type Interaction Effects

6.2.1 6.2.1 Same-Turn Alignment by LLM Type

```
## === Same-Turn Alignment: LLM Type Interaction ===

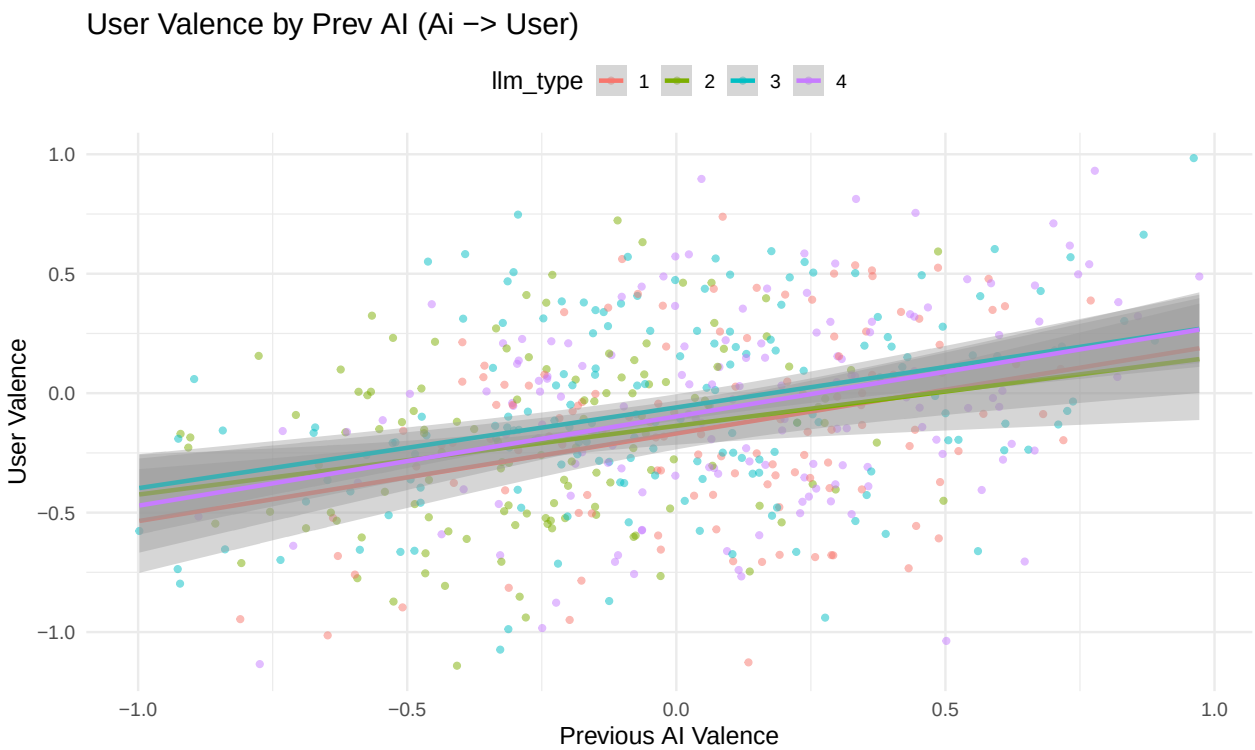
##                                Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                      0.1189     0.0442  73.4467  2.6905   0.0088
## user_valence                     0.2647     0.0706 636.6685  3.7504   0.0002
## factor(llm_type)2               -0.3213     0.0633  76.9204 -5.0768   0.0000
## factor(llm_type)3               -0.1115     0.0581  71.7833 -1.9200   0.0588
## factor(llm_type)4                0.0218     0.0597  71.3172  0.3654   0.7159
## user_valence:factor(llm_type)2  -0.0995     0.1017 639.9991 -0.9780   0.3284
## user_valence:factor(llm_type)3   0.1356     0.0967 628.5873  1.4034   0.1610
## user_valence:factor(llm_type)4  -0.0211     0.0960 638.1429 -0.2200   0.8260
```



6.2.2 6.2.2 Previous-Turn Alignment by LLM Type

```
## === Previous-Turn Alignment: LLM Type Interaction ===

##                                  Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                       -0.1609     0.0495  68.9160 -3.2500   0.0018
## ai_valence_lag1                    0.2538     0.1022 536.1075  2.4830   0.0133
## factor(llm_type)2                  0.0054     0.0741  84.7923  0.0723   0.9425
## factor(llm_type)3                  0.0984     0.0652  67.9910  1.5094   0.1358
## factor(llm_type)4                  0.0784     0.0679  70.3908  1.1550   0.2520
## ai_valence_lag1:factor(llm_type)2 -0.0441     0.1471 558.2680 -0.2998   0.7645
## ai_valence_lag1:factor(llm_type)3 -0.0684     0.1255 532.6648 -0.5446   0.5862
## ai_valence_lag1:factor(llm_type)4 -0.0056     0.1327 542.1418 -0.0421   0.9664
```



6.2.3 6.2.3 Per-Story Correlations by LLM Type

```
## === Same-Turn Correlations by LLM Type ===
```

```
## # A tibble: 4 x 4
##   llm_type mean_rho sd_rho n_stories
##      <int>    <dbl>  <dbl>     <int>
## 1        1    0.279  0.383        18
## 2        2    0.192  0.383        18
## 3        3    0.228  0.394        24
## 4        4    0.151  0.390        21
```

```
##
## ANOVA (LLM type effect on same-turn rho):
```

```
##                  Df Sum Sq Mean Sq F value Pr(>F)
## factor(llm_type)  3  0.173 0.05765   0.382  0.766
## Residuals        77 11.617 0.15087
```

6.3 6.3 Valence Distribution by Conditions

6.3.1 6.3.1 Valence by Starter

```
## # A tibble: 4 x 4
##   starter agent        mean_valence sd_valence
##   <chr>   <chr>               <dbl>      <dbl>
## 1 ai      ai_valence         0.0136      0.381
## 2 ai      user_valence      -0.0878      0.384
## 3 user    ai_valence        -0.0214      0.401
## 4 user    user_valence      -0.106       0.410
```

6.3.2 6.3.2 Valence by LLM Type

```
## # A tibble: 8 x 5
##   llm_type agent        mean_valence sd_valence n_turns
##      <int> <chr>               <dbl>      <dbl>   <int>
## 1        1 ai_valence         0.0854      0.344     144
## 2        1 user_valence      -0.127       0.406     144
## 3        2 ai_valence        -0.232       0.298     144
## 4        2 user_valence      -0.182       0.379     144
## 5        3 ai_valence        -0.0166      0.421     192
## 6        3 user_valence      -0.0600      0.387     192
## 7        4 ai_valence         0.131       0.377     168
## 8        4 user_valence      -0.0411      0.404     168
```

6.4 7. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] here_1.0.2      afex_1.4-1      cowplot_1.1.3   patchwork_1.3.0
##  [5] zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1  lmerTest_3.1-3
##  [9] lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3 forcats_1.0.0
## [13] stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2     readr_2.1.5
## [17] tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           RColorBrewer_1.1-3  plyr_1.8.9
## [22] abind_1.4-8         multcomp_1.4-26     withr_3.0.2
## [25] numDeriv_2016.8-1.1 grid_4.4.2          fansi_1.0.6
## [28] xtable_1.8-4        colorspace_2.1-1    emmeans_2.0.0
## [31] scales_1.3.0        MASS_7.3-61         tinytex_0.52
## [34] cli_3.6.3           mvtnorm_1.3-3       rmarkdown_2.28
## [37] crayon_1.5.3        ragg_1.3.2          generics_0.1.3
## [40] rstudioapi_0.16.0   reshape2_1.4.4      tzdb_0.4.0
```

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## [43] minqa_1.2.8         splines_4.4.2       assertthat_0.2.1
## [46] parallel_4.4.2      vctrs_0.6.5         boot_1.3-31
## [49] sandwich_3.1-1      carData_3.0-5       car_3.1-3
## [52] hms_1.1.3           bit64_4.0.5         Formula_1.2-5
## [55] systemfonts_1.1.0   glue_1.7.0          nloptr_2.1.1
## [58] codetools_0.2-20    gtable_0.3.5        munsell_0.5.1
## [61] pillar_1.9.0        htmltools_0.5.8.1   R6_2.5.1
## [64] textshaping_0.4.0   rprojroot_2.1.1     vroom_1.6.5
## [67] evaluate_1.0.3      lattice_0.22-6      Rcpp_1.0.13
## [70] coda_0.19-4.1       nlme_3.1-166        mgcv_1.9-1
## [73] xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.